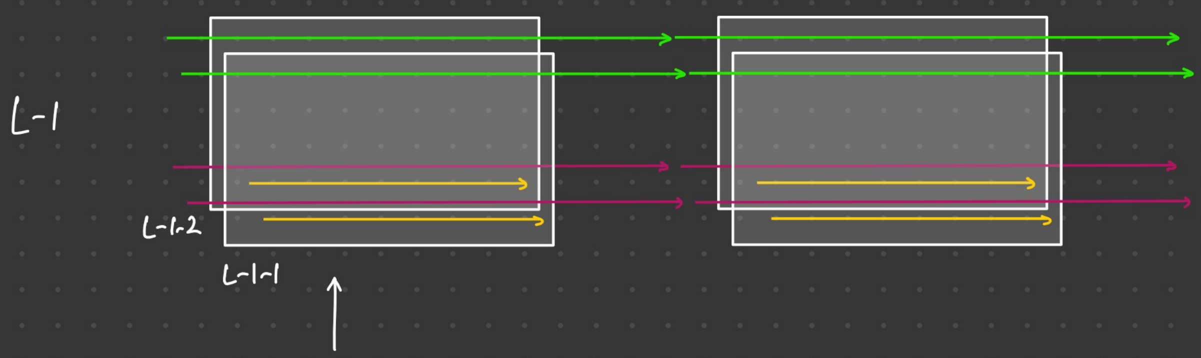
Agenda - 22-02-2025

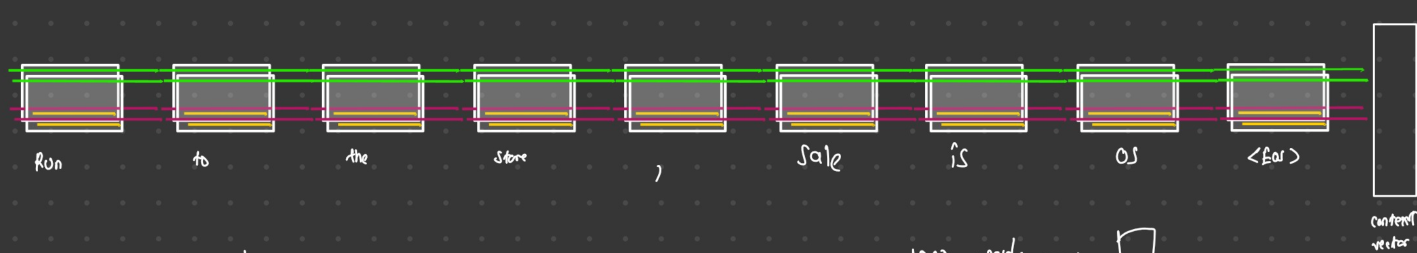
- Attention
- Transformer
  - Single head attention
  - Multi head attention



Run to the store, sale is on <EOS>

T→ 0 1 2 3 4 5 6 7 8

ANN → 2
LSTM → 50 - 100



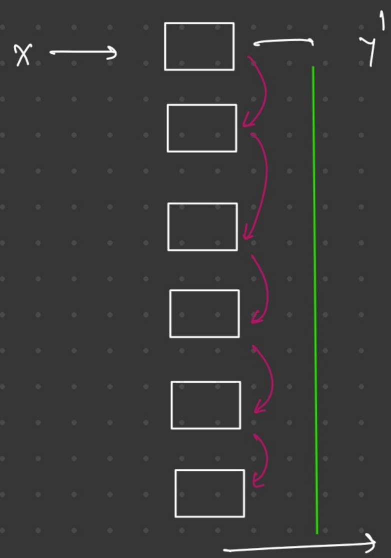
→ 1000 words →

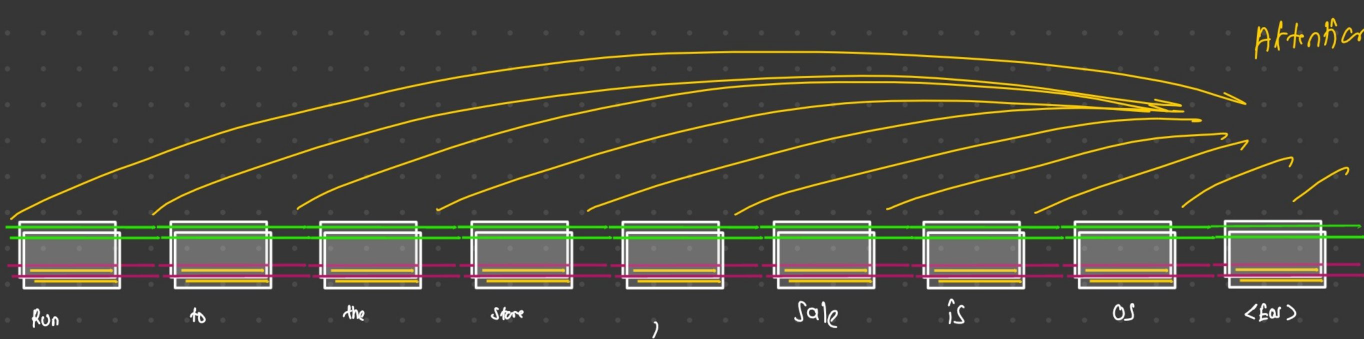
~~Run~~ ~~To~~ The store, sale is on

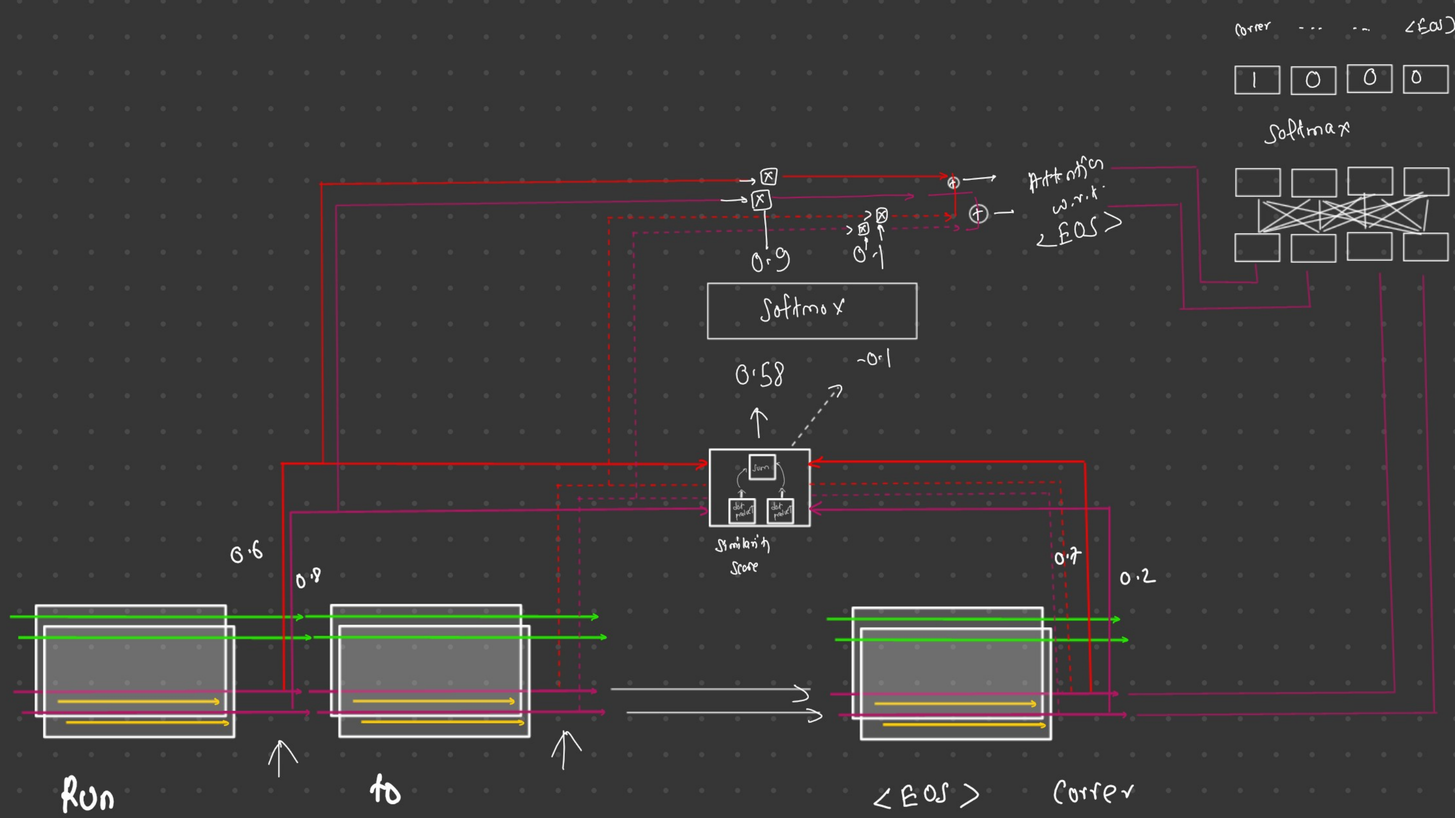
RNN

→ Hidden state – short term memory

→ Cell state – long term memory







Cosine similarity :

$$\frac{\sum_{i=1}^{n} A_i B_i}{\cancel{\sqrt{\sum_{i=1}^{n} A_i^2} \sqrt{\sum_{i=1}^{n} B_i^2}}}$$

$$\begin{array}{lcl} & & \quad A_1 \quad\; A_2 \\ \text{Run} & \leftarrow A \rightarrow & [0.6 \,,\, 0.8] \\ \langle EOS \rangle & \leftarrow B \rightarrow & [0.7 \,,\, 0.2] \\ & & \quad B_1 \quad\; B_2 \end{array}$$

$$(0.6 \times 0.7) + (0.8 \times 0.2) \rightarrow \boxed{.58}$$

Attention $\longrightarrow$ sentence w.r.t your output token

Context vector

$\longrightarrow$ encoded vect

Research paper :— Attention is all you need

— Transformers

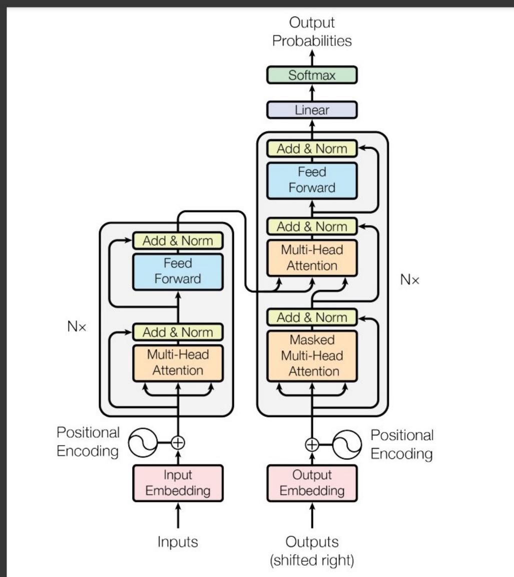
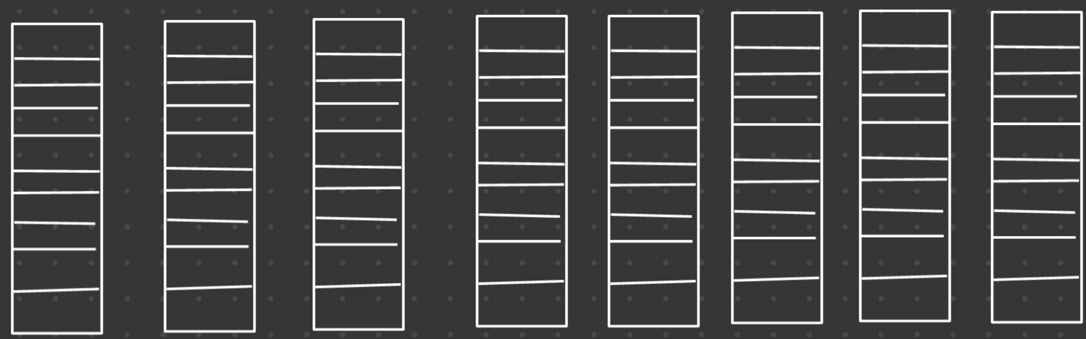


Figure 1: The Transformer - model architecture.

(1) Calculate Embedding

Run to the store , Sale is on.



10 dim embedding

$d_{model} = 10$

(2) positional Encoding

$$PE(pos, 2i) = \sin \frac{pos}{10000^{\frac{2i}{d_{model}}}}$$

$$PE(pos, 2i+1) = \cos \frac{pos}{10000^{\frac{2i}{d_{model}}}}$$

pos - position index of the word in the sequenece

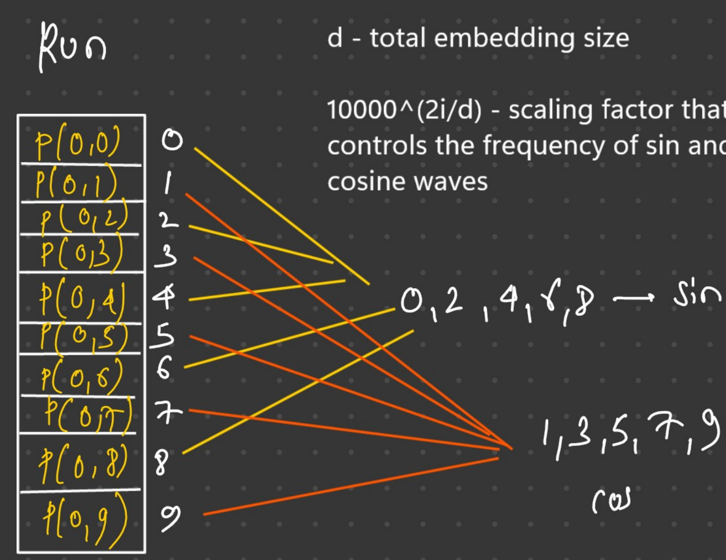
i - index of the embeddings dimension

d - total embedding size

10000^(2i/d) - scaling factor that controls the frequency of sin and cosine waves

$i = 0$

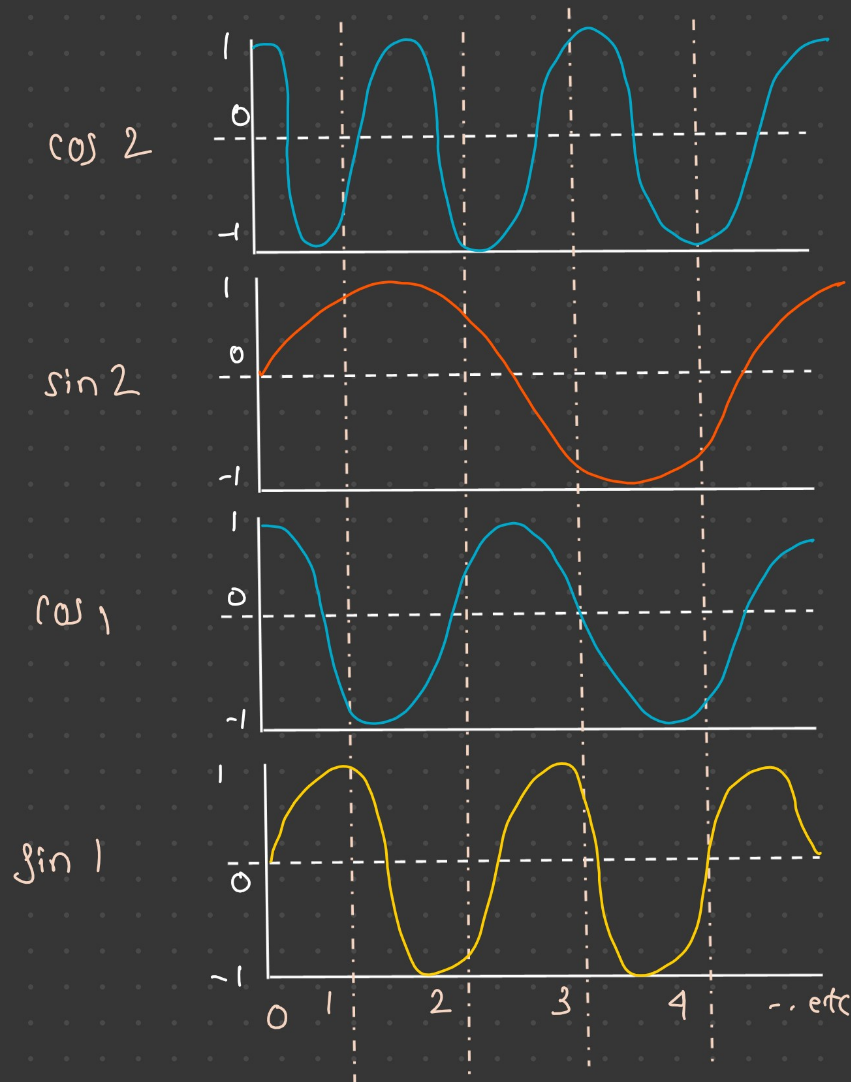
$$PE(pos, 0) = \sin \frac{0}{10000^{\frac{2 \times 0}{10}}} = \leftarrow$$



$i=0, \quad 10000^{0} = 1$

$i=1, \quad 10000^{0.2} = x$

$i=2, \quad 10000^{0.4} =$



p(0,0)	$\xrightarrow{\sin}$	0
p(0,1)	$\xrightarrow{\cos}$	1
p(0,2)	$\xrightarrow{\sin}$	0
p(0,3)	$\rightarrow$	1
p(0,4)	$\rightarrow$	0
p(0,5)	$\rightarrow$	1
p(0,6)	$\rightarrow$	0
p(0,7)	$\rightarrow$	1
p(0,8)	$\rightarrow$	0
p(0,9)	$\rightarrow$	1

p(1,0)	$\xrightarrow{\sin}$	1
p(1,1)	$\xrightarrow{\cos}$	-0.9
p(1,2)	$\xrightarrow{\sin}$	0.7
p(1,3)	$\rightarrow$	-0.6
p(1,4)	$\rightarrow$	x
p(1,5)	$\rightarrow$	x
p(1,6)	$\rightarrow$	x
p(1,7)	$\rightarrow$	x
p(1,8)	$\rightarrow$	x
p(1,9)	$\rightarrow$	x

→ Embedding is of shape 10

sin → 5

cos → 5

$\boxed{512} \rightarrow$

sin $512/2$

cos $512/2$

Embedding	positional information	positional embedding
0.9	P(0,0)	0.9
0.5	P(0,1)	0.5
0.4	P(0,2)	0.4
0.2	P(0,3)	0.2
0.1	P(0,4)	0.1
0.35	P(0,5)	0.35
0.7	P(0,6)	0.7
0.8	P(0,7)	0.8
0.96	P(0,8)	0.96
0.22	P(0,9)	0.22

Embedding + positional information → positional embedding

sentence — run to store , - - - -

position ← 0 1 2

sin —
cos —

Hi , How are you

↓

← position

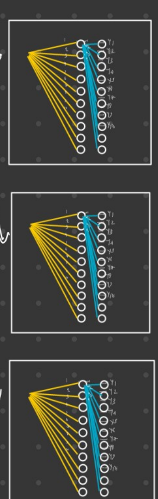
embedding + posin

—— po

[10 x 10]

run
to
the
store
,
sale
is
on
.

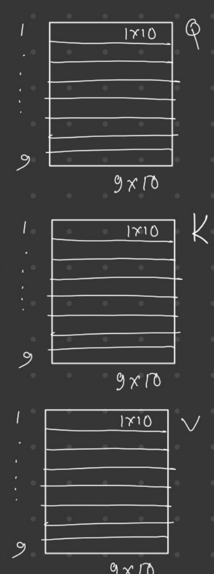
[9x10]

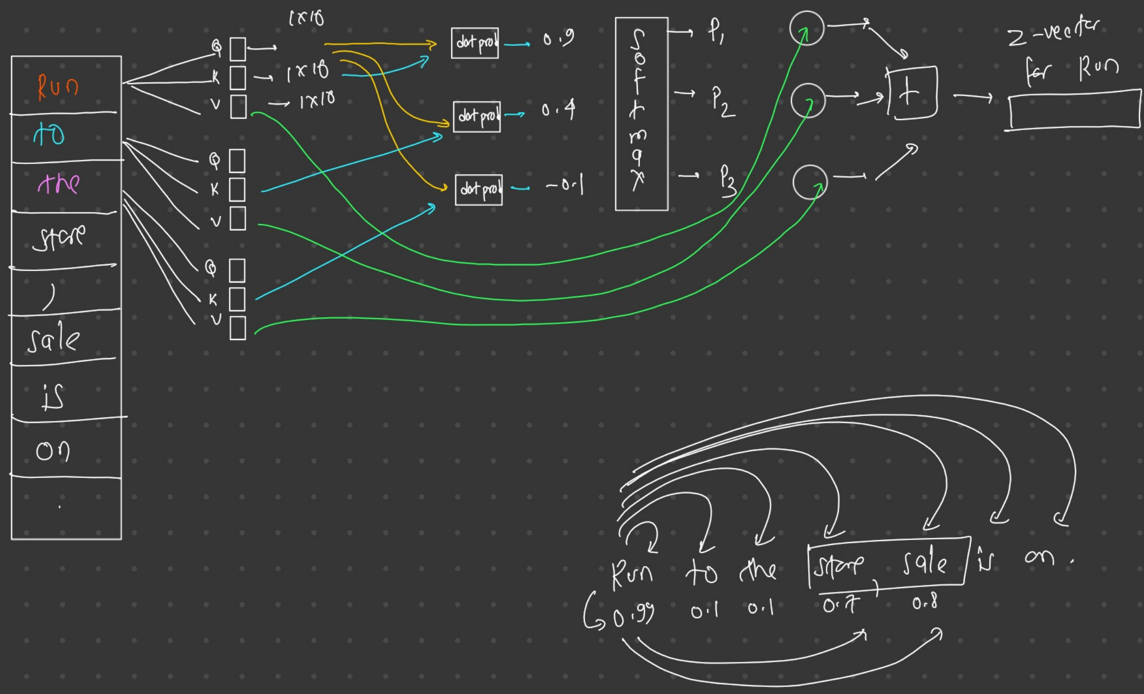


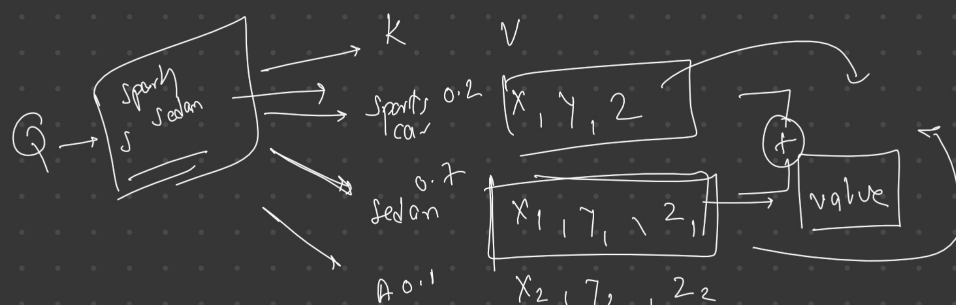
Q K V

Q K V

Q K V







self attn

Q → information

K → information

V → information

Q' & $K_{1..n}$

↓

similarity w.r.t Q'

↓

softmax

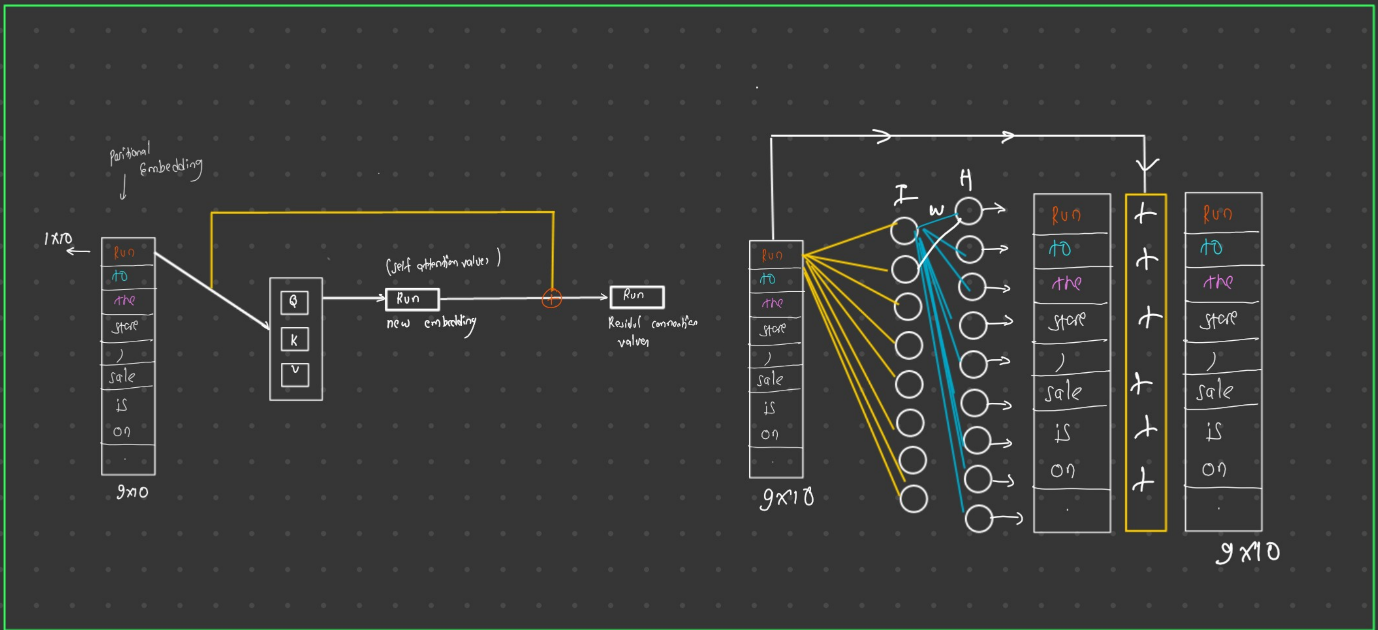
↓

weightage of those keys w.r.t. Q

Scaling

↓

softmax → $\boxed{V}$ →

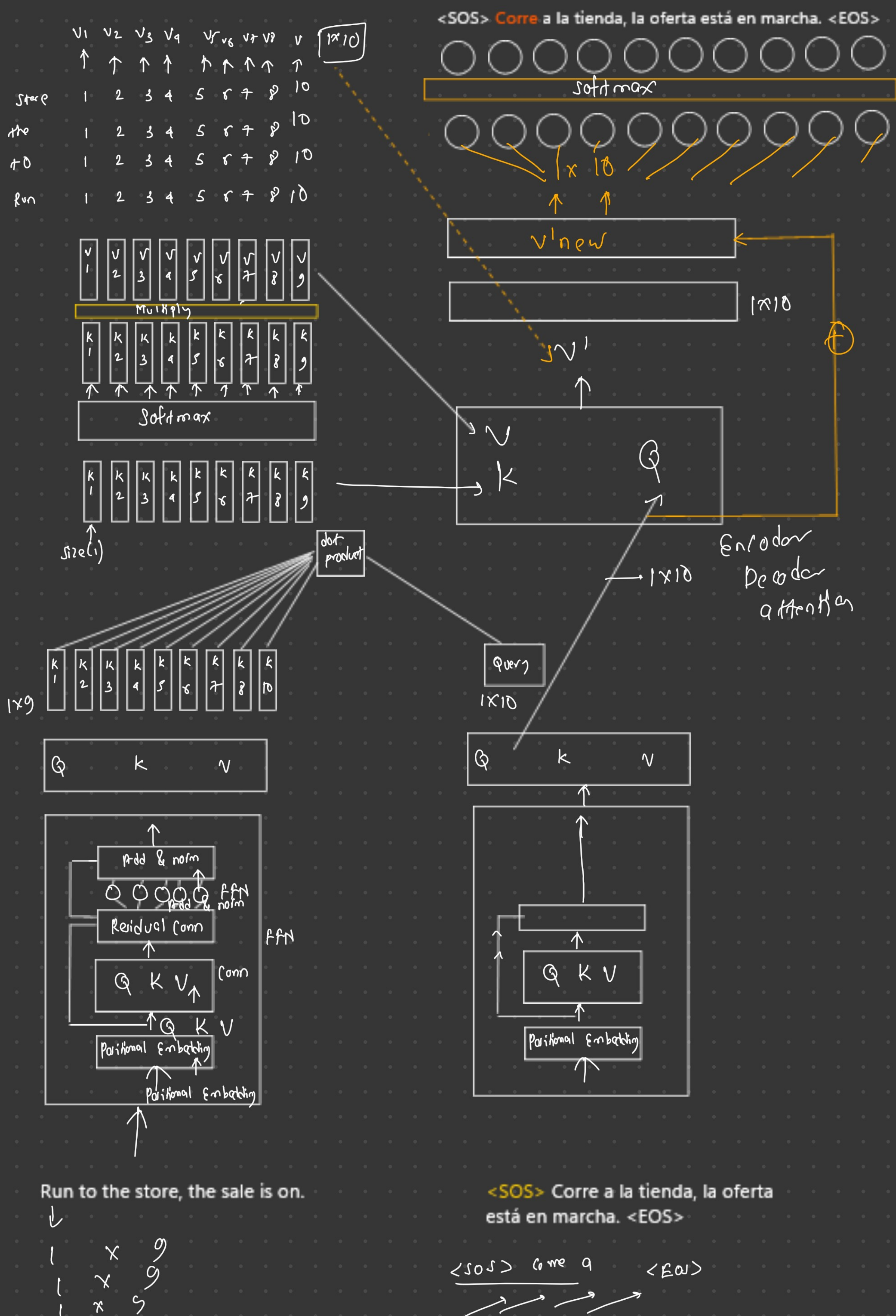


9x10 →

Encoder

Run to the store, the sale is on.

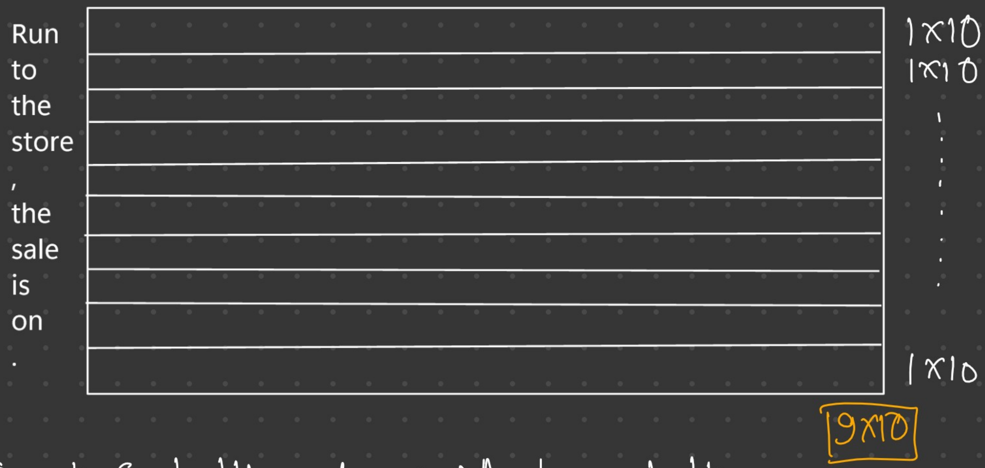
Corre a la tienda, la oferta está en marcha.



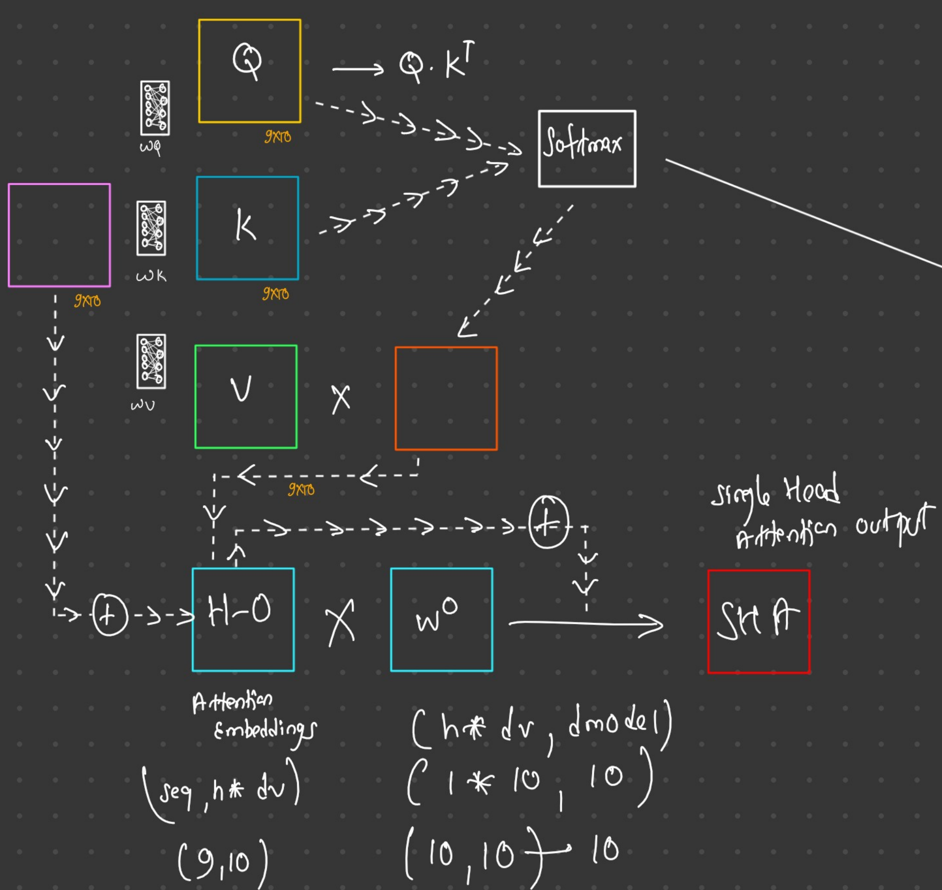
<SOS> Corre a la tienda, la oferta está en marcha. <EOS>
Softmax
1x10
v'new
1x10
V'
Encoder Decoder attention
Multiply
Softmax
size(1)
dot product
Query
1x10
1x9
Q k V
Add & norm
FFN
Residual conn
Q K V
Conn
Positional Embedding
Run to the store, the sale is on.
1 x 9
<SOS> Corre a la tienda, la oferta está en marcha. <EOS>
<SOS> Corre a <EOS>

	Run	to	the	store	,	the	sale	is	on	.
Run										
to										
the										
store										
,										
the										
sale										
is										
on										
.										

Run	
to	
the	
store	
,	
the	
sale	
is	
on	
.	



Input Embedding + positional embedding



$$\text{Softmax}\left(\frac{QK^T}{\sqrt{dk}}\right)$$

$dk = 10$

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{dk}}\right)V$$

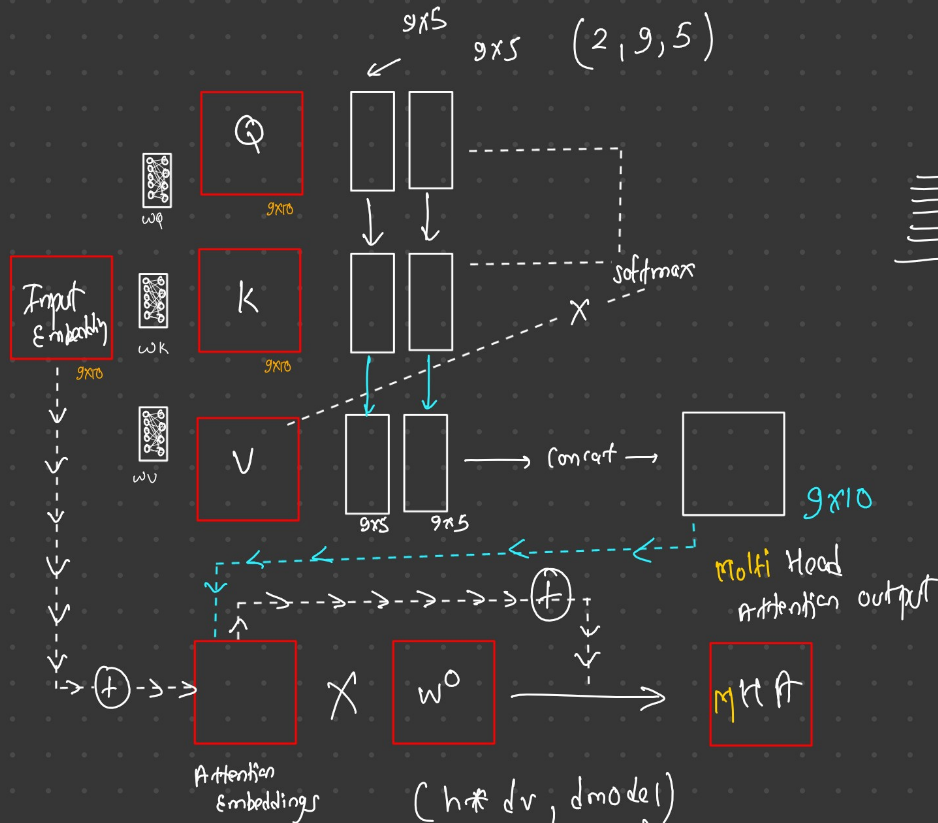
10 ← dmodel → size of embedding vector

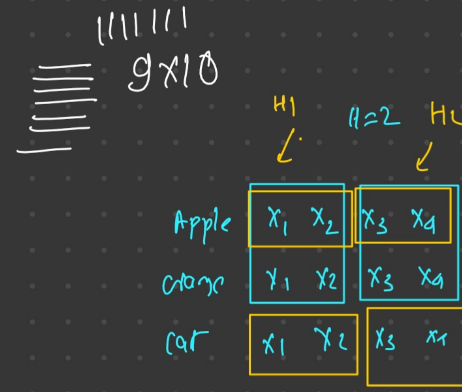
size → sequence lenght → 9

1 ← h → no of heads

10 ← $dk = dv = dmodel/h$

Head → 2





(seq, h* dv)

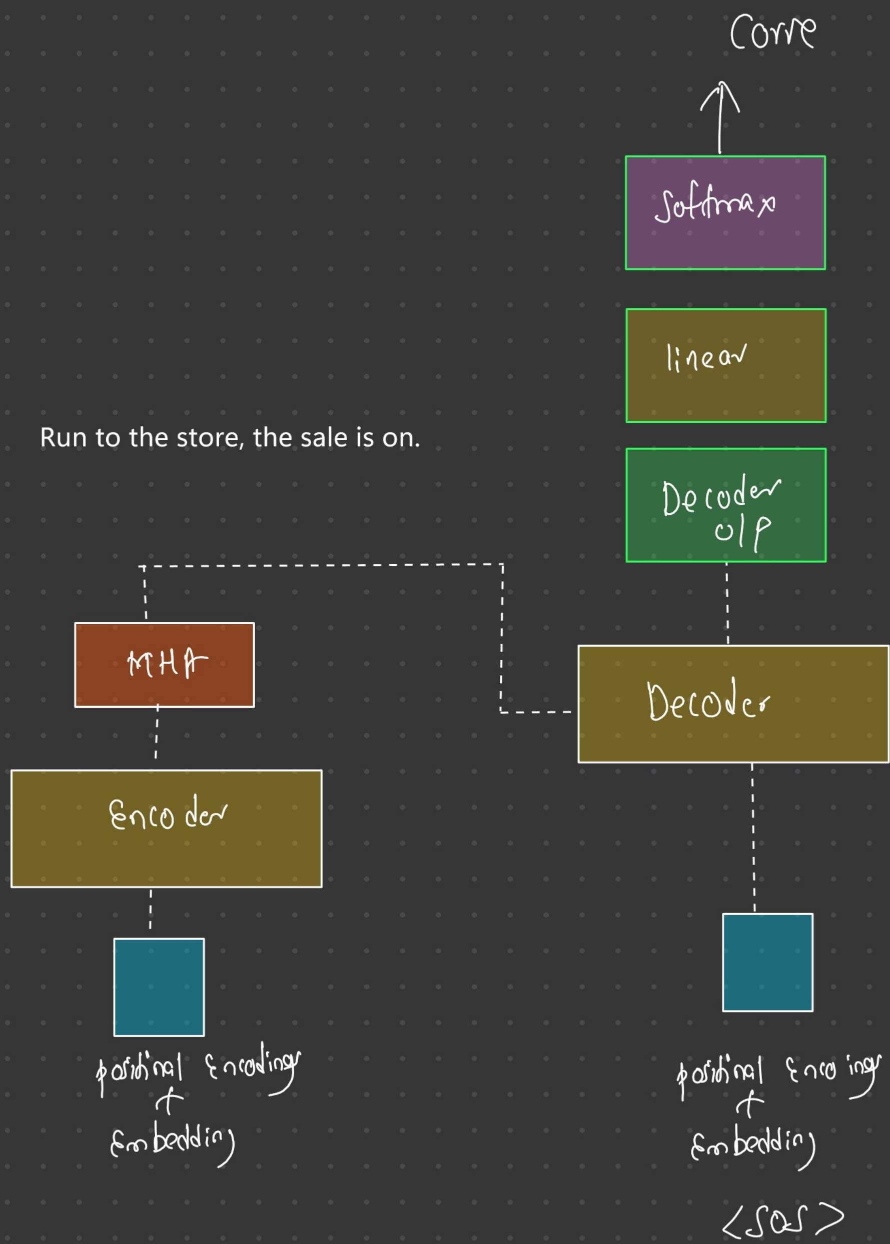
(9

(h* dv, dmodel)

(1 * 10, 10)

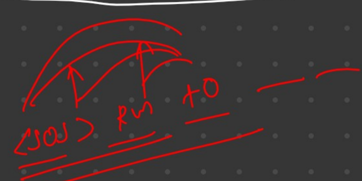
(10 10) → 10

Run to the store, the sale is on.

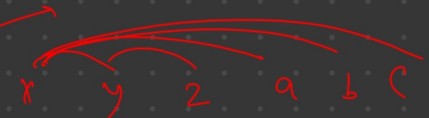


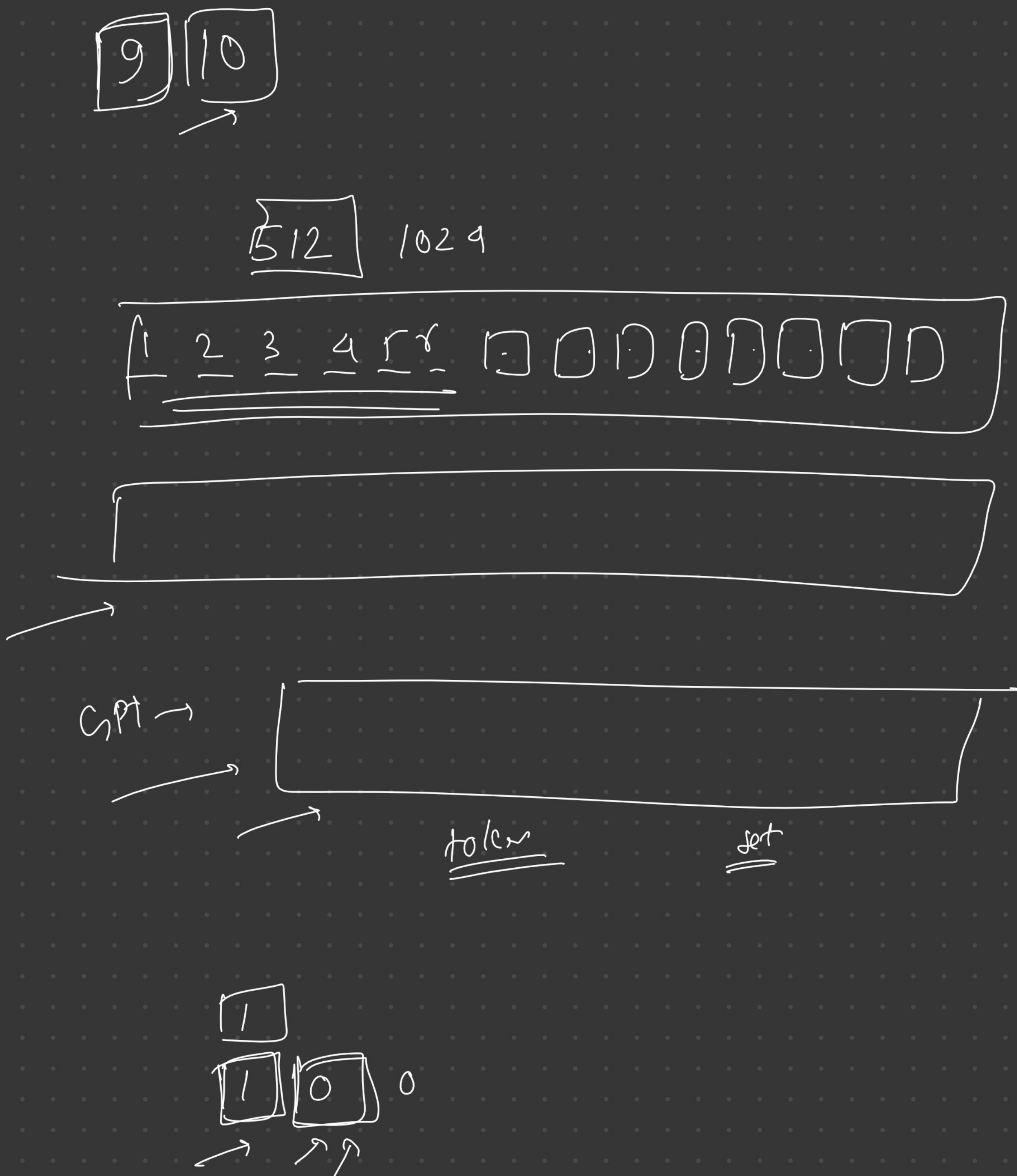
<SOS> Run to the store, the sale is on. <EOS>

Masked Multi Head attention.



	Run	to	the	store	,	the	sale	is	on	.
Run	x	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$
to										
the										
store										
,										
the										
sale										
is										
on										
.										




9
10
512
1024
1 2 3 4 5 5
GPT →
token
set
1
1 0 0